

Hetero-interpenetrated metal–organic frameworks

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Interpenetrated metal–organic frameworks (MOFs) comprise two or more lattices that are mutually entangled. Interpenetration can be used to tune the structures and pore architectures of MOFs to influence, for example, their stability or interactions with guest molecules. The interpenetrating sublattices are typically identical, but hetero-interpenetrated MOFs, which consist of sublattices that are different from one another, have also been serendipitously produced. Here we describe a strategy for the deliberate synthesis of hetero-interpenetrated MOFs. We use the cubic α -MUF-9 framework as a host sublattice to template the growth of a second sublattice within its pores. Three different secondary sublattices are grown—two of which are not known as standalone MOFs—leading to three different hetero-interpenetrated MOFs. This strategy may serve to combine different properties into one material. We produce an asymmetric catalysis by allocating separate roles to the interpenetrating sublattices in a hetero-interpenetrated MOF: an achiral secondary amine on one sublattice provides the catalytic activity, while the chiral α -MUF-10 host imparts asymmetry to aldol and Henry reactions.

Interpenetration^{1,2} is a common phenomenon in MOFs, where entangled, interlocked sublattices exist in the same crystal. Methods to control interpenetration are important since interpenetration governs the size and chemical environment of the pores, the diffusivity of guest molecules and the framework stability^{3–8}. The interpenetrating sublattices are typically identical, but hetero-interpenetrated MOFs comprise sublattices that are different from one another. Such frameworks are fascinating from the design and structural viewpoints, and they may exhibit properties that are not observed in the individual lattices on their own. For example, bringing together donor groups on one network with acceptor groups on another may result in new optical, magnetic and/or electronic characteristics. Complementary functional groups may also be precisely placed in the pores of hetero-interpenetrated MOFs to deliver unique adsorption, catalysis and sensing applications. Although occasional examples of hetero-interpenetrated MOFs have been discovered serendipitously^{3,9–12}, their deliberate synthesis remains elusive. Recent computational studies have identified sublattices that are mutually compatible *in silico*. However, these hetero-interpenetrated

MOFs have not been experimentally realized^{13,14}. Under conventional synthetic protocols, one-pot methods using structurally related ligands will typically produce multivariate frameworks¹⁵, while ligands with different geometries produce either multicomponent MOFs¹⁶ or mixed phases. In this light, the deliberate synthesis of hetero-interpenetrated MOFs is not straightforward.

To address this challenge, we formulated a two-step strategy towards the deliberate synthesis of hetero-interpenetrated MOFs. In this approach, a non-interpenetrated MOF acts as a host sublattice. A second, interpenetrating framework is grown in the pore space of this host by subsequent secondary growth. While the interpenetrating sublattice must be geometrically compatible with the host, it can be chemically distinct to deliver a hetero-interpenetrated framework. There is potential for this strategy to be disrupted by competing processes. For example, the second framework may grow outside the pores of the first in a separate phase, the incoming components may displace those of the original framework or the growth of the second framework may be hindered by mass transfer requirements. We herein

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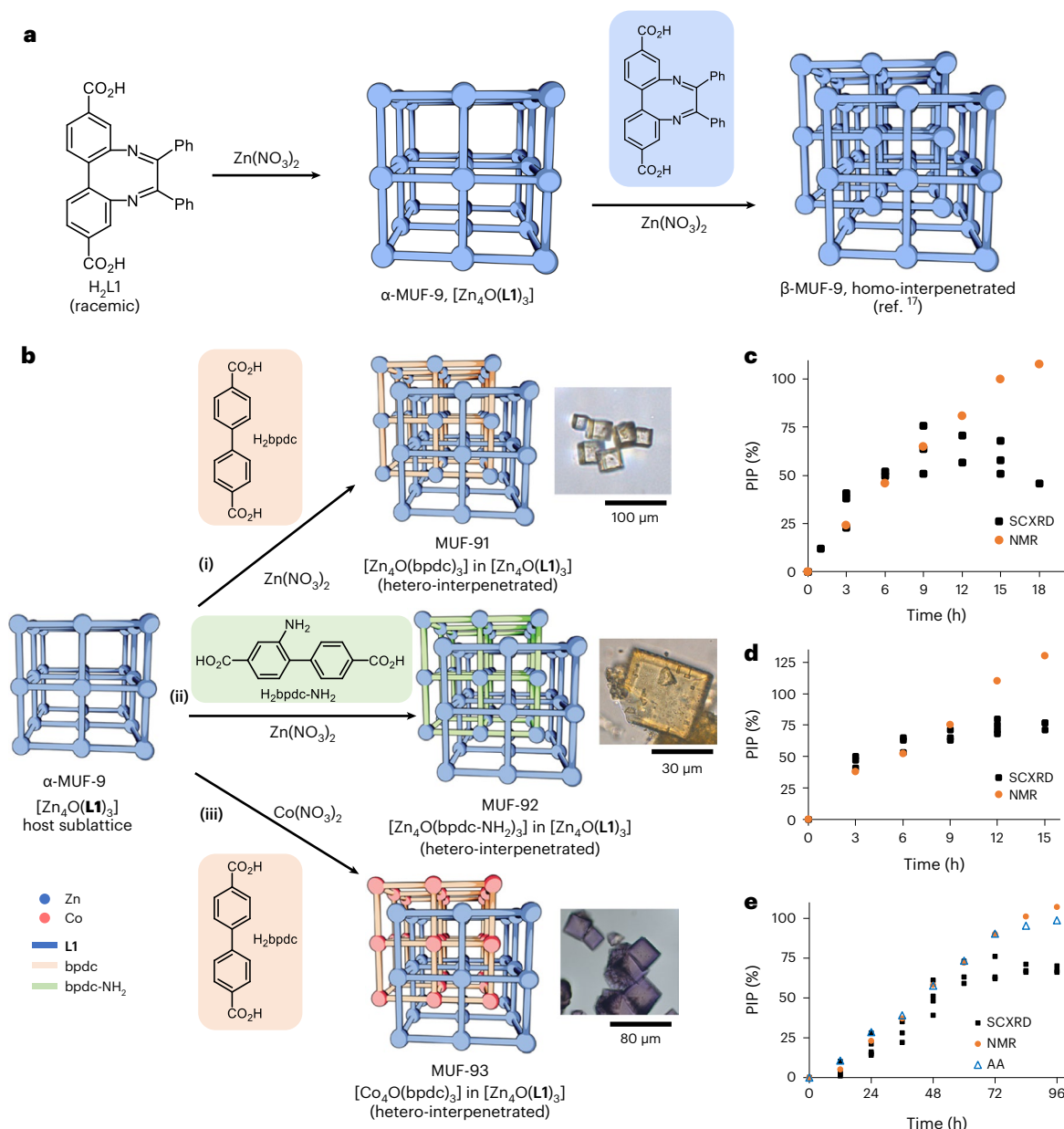


Fig. 1 | Hetero-interpenetrated versus homo-interpenetrated MOFs.

a, Non-interpenetrated α -MUF-9 can serve as a host for the growth of another copy of the $[\text{Zn}_4\text{O}(\text{L1})_3]$ lattice to make the homo-interpenetrated MOF β -MUF-9. **b**, The experimental strategy towards the hetero-interpenetrated MOFs MUF-91 (i), MUF-92 (ii) and MUF-93 (iii), where α -MUF-9 templates a second sublattice that is chemically distinct from itself. Optical micrographs of the MUF-91, MUF-92 and MUF-93 are presented. **c–e**, The PIP level as a function of time as deduced from various experimental techniques for MUF-91 (c), MUF-92 (d) and

MUF-93 (e). Black squares represent the PIP (fractional interpenetration) of multiple individual crystals as determined by SCXRD. Orange circles represent the PIP determined by ^1H NMR spectroscopy. Blue triangles represent the ratio of zinc(II) to cobalt(II) in MUF-93 as determined by AA spectroscopy. The best measure of the PIP is given by SCXRD since the growth of a shell layer adversely influences the PIP determined from ^1H NMR and AA spectroscopic measurements at longer reaction times. Trendlines and confidence intervals are presented in Supplementary Figs. 2, 7 and 14.

present a successful realization of this strategy that circumvents these potential limitations.

We started from α -MUF-9 (MUF, Massey University Framework), which is a cubic MOF with a primitive cubic (pcu) lattice built up from Zn_4O nodes and linear dicarboxylate linkers (**L1**; Fig. 1). Non-interpenetrated MUF-9 (denoted as α -MUF-9) can be synthesized in a bulky solvent such as *N,N*-di-*n*-butylformamide (DBF). In earlier work, we found that α -MUF-9 promotes the growth of a second $[\text{Zn}_4\text{O}(\text{L1})_3]$ lattice via favourable non-covalent interactions to produce β -MUF-9, a conventional homo-interpenetrated MOF¹⁷. We speculated that secondary growth could instead involve sublattices with different compositions but the same metric parameters (Fig. 1b).

This was achieved using zinc(II) or cobalt(II) metal ions to build up the M_4O clusters and ligands based on biphenyl-4,4'-dicarboxylate and its close analogues.

The concept of partial interpenetration is central to this work. In most interpenetrated MOFs, the number of interpenetrating lattices is an integer. However, in partially interpenetrated MOFs, different regions of the crystal are composed of different numbers of sublattices^{17–20}. This results in a fractional value for the overall level of partial interpenetration, which can be expressed as a percentage (PIP). For a doubly interpenetrated MOF, the PIP corresponds to the occupancy of the second sublattice. In some reported cases, the PIP level is fixed^{18,19}, while in others it can be controlled^{17,20}.

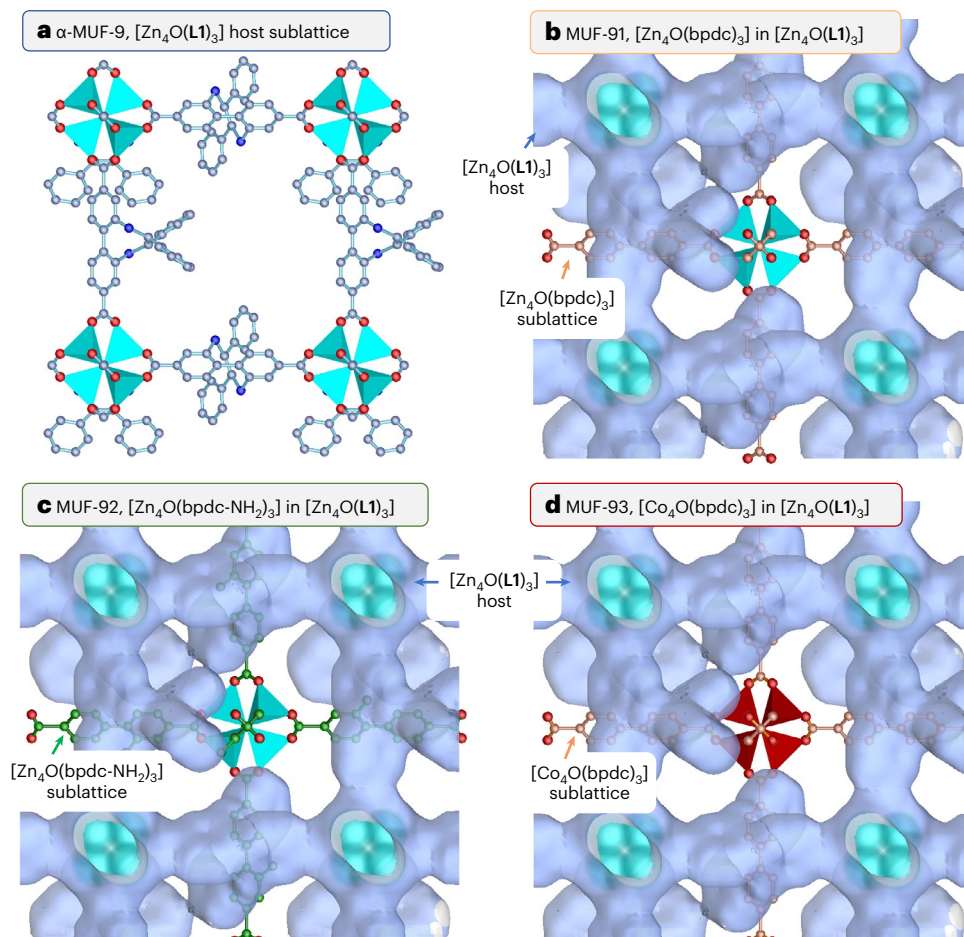


Fig. 2 | The structures of α -MUF-9 and MUF-91–MUF-93 determined by SCXRD. a, The α -MUF-9 host sublattice showing representative positions of the symmetry-disordered **L1** ligands. **b–d**, The structures of MUF-91 (**b**), MUF-92 (**c**) and MUF-93 (**d**), with the van der Waals surface of α -MUF-9 shown in blue to highlight the space available for growth of the interpenetrating sublattice

and the scope for attractive non-covalent interactions. The $[\text{Zn}_4\text{O}(\text{bpdc})_3]$, $[\text{Zn}_4\text{O}(\text{bpdc-NH}_2)_3]$ and $[\text{Co}_4\text{O}(\text{bpdc})_3]$ sublattices in MUF-91–MUF-93 are shown with ball-and-stick models. Hydrogen atoms are omitted from all figures for clarity.

Results and discussion

The hetero-interpenetrated MUF-91 framework

Our first target was MUF-91, in which a $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ sublattice interpenetrates α -MUF-9 ($[\text{Zn}_4\text{O}(\text{L1})_3]$; Fig. 1b(i)). We first prepared α -MUF-9 using racemic **H₂L1** and then incubated it in a secondary growth solution comprising 4,4'-biphenyldicarboxylic acid (**H₂bpdc**) and $\text{Zn}(\text{NO}_3)_2$ in DBF. We found 2-fluorobenzoic acid (FBA) to be a useful additive to suppress the formation of a standalone phase of $[\text{Zn}_4\text{O}(\text{bpdc})_3]$. In addition, the cloudy supernatant was removed periodically throughout the secondary growth process to eliminate small crystallites that self-nucleate from **H₂bpdc** and $\text{Zn}(\text{NO}_3)_2$. The PIP level of MUF-91, which equates with the occupancy of the interpenetrating $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ sublattice, increases over time and can be determined quantitatively by single-crystal X-ray diffraction (SCXRD), powder X-ray diffraction (PXRD) and ^1H NMR spectroscopy (Fig. 1c). A crystallographic model was developed for the SCXRD datasets (Fig. 2b and Supplementary Tables 1 and 2). The occupancy of the secondary $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ sublattice was refined as a free variable to give a direct measure of the PIP level over the course of secondary growth. Restraints were used to ensure refinement stability and chemical correctness, and these were loosened as far as the data quality would allow. The $P\bar{4}3m$ space group of α -MUF-9 is consistently maintained. After 9 hours, the PIP level reaches 70%, which is the upper limit for the occupancy of the $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ sublattice (Fig. 1c). We could also gauge the PIP level

over time by quantifying the changes in the PXRD patterns, and excellent agreement with the SCXRD data was observed. The intensity of the PXRD peak at $2\theta = 5.2^\circ$ decreases in intensity, and the peak at 7.3° increases (Supplementary Fig. 1). These results correspond to the (100) and (110) reflections that are, respectively, extinguished and enhanced by the growing $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ sublattice. These changes mirror those observed during the homo-interpenetration of MUF-9 (ref. 17). In accord with the diffraction data, the ^1H NMR spectroscopic analysis of digested MUF-91 samples shows that the amount of bpdc relative to **L1** increases with time. The PIP level deduced from the bpdc/**L1** ratio measured by ^1H NMR spectroscopy on digested samples matches the PIP level given by the diffraction data for the first 9 hours of secondary growth (Fig. 1c). This confirms that bpdc does not simply displace **L1** from the $[\text{Zn}_4\text{O}(\text{L1})_3]$ host sublattice. After 9 hours, a shell of a different phase forms on the surface of the MUF-91 crystals, which is clearly distinguishable by PXRD (Supplementary Fig. 1) and optical microscopy (Supplementary Fig. 3). The growth of this $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ shell is only possible when secondary growth has occurred to a substantial extent and the occupancy of the interpenetrating lattice is already high. After this point, the bpdc/**L1** ratio measured by ^1H NMR spectroscopy increases beyond the PIP level since the shell is composed of $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ with no **L1**. To optimize the occupancy of the interpenetrating lattice, we prepared MUF-91 starting from microcrystalline α -MUF-9 rather than large single crystals (Supplementary Fig. 6).

PIP levels of ~75% were obtained since the smaller particle size allows for more rapid mass transport during secondary growth. As a control experiment, α -MUF-9 was incubated with H_2bpdc and FBA in DBF. Over a period of 9 hours, only 4% of the **L1** linkers in MUF-9 were displaced by bpdc (Supplementary Fig. 28), which places an upper bound on the exchange that can occur during the synthesis of MUF-91. The well-controlled formation of hetero-interpenetrated MUF-91 via secondary growth contrasts with the direct reaction of $\text{H}_2\text{L1}$, H_2bpdc and $\text{Zn}(\text{NO}_3)_2$ in *N,N*-diethylformamide. This reaction produces a cubic, doubly interpenetrated framework with the **L1** and bpdc linkers distributed randomly throughout the two sublattices, that is, a multivariate, homo-interpenetrated framework. The driving force for the growth of the $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ sublattice in the pores of α -MUF-9 (Fig. 2a) is the attractive non-covalent interactions that are enabled when the two networks lie in close proximity (Fig. 2b)¹⁷. The templating ability of α -MUF-9 allows the hetero-interpenetrated MUF-91 to grow while the formation of a standalone $[\text{Zn}_4\text{O}(\text{bpdc})_3]$ framework is inhibited by the 2-FBA modulator.

The hetero-interpenetrated MUF-92 framework

MUF-92 comprises $[\text{Zn}_4\text{O}(\text{bpdc-NH}_2)_3]$ ($\text{bpdc-NH}_2 = 2\text{-amino-}4,4'\text{-biphenyldicarboxylate}$) interpenetrated in α -MUF-9 (Fig. 1b(ii)). $[\text{Zn}_4\text{O}(\text{bpdc-NH}_2)_3]$ is not known to form as a standalone framework⁷, but it is observed here courtesy of the templating effect of the $[\text{Zn}_4\text{O}(\text{L1})_3]$ host sublattice. The secondary growth conditions that produce MUF-92 employ α -MUF-9 crystals together with $\text{H}_2\text{bpdc-NH}_2$, $\text{Zn}(\text{NO}_3)_2$ and the FBA modulator in DBF. We monitored the growth of the $[\text{Zn}_4\text{O}(\text{bpdc-NH}_2)_3]$ sublattice by SCXRD, PXRD and ^1H NMR spectroscopy (Fig. 1d, Supplementary Figs. 7 and 8 and Supplementary Table 3). The $[\text{Zn}_4\text{O}(\text{bpdc-NH}_2)_3]$ sublattice grows in over an incubation period of 9 hours to reach a PIP level of 74%. The phenyl rings of the bpdc- NH_2 ligand adopt an orthogonal conformation, which is impossible for **L1** and thus allows the two sublattices in MUF-92 to be unambiguously differentiated by SCXRD (Fig. 2c). This indicates there is little, if any, displacement of the linkers from the $[\text{Zn}_4\text{O}(\text{L1})_3]$ sublattice by bpdc- NH_2 , which was additionally verified by a control experiment (Supplementary Fig. 30). Beyond 9 hours, a shell layer comprising Zn(II) and bpdc- NH_2 grows around the crystals, which means the bpdc- NH_2 /**L1** ratio deduced by NMR spectroscopy continues to rise while the PIP level of the MUF-92 core remains constant (Supplementary Fig. 9). When microcrystals of α -MUF-9 are used for secondary growth, the occupancy level of the $[\text{Zn}_4\text{O}(\text{bpdc-NH}_2)_3]$ sublattice in MUF-92 reaches ~73% before it is inhibited by the Zn(II)/bpdc- NH_2 shell layer (Supplementary Figs. 11 and 12).

The hetero-interpenetrated MUF-93 framework

MUF-93 features α -MUF-9 interpenetrated by $[\text{Co}_4\text{O}(\text{bpdc})_3]$ (Fig. 1b(iii)). $[\text{Co}_4\text{O}(\text{bpdc})_3]$ is not known as a standalone framework²¹, although $[\text{Co}_4\text{O}(\text{1,4-benzenedicarboxylate})_3]$ has been reported starting from a preformed Co_4OL_6 cluster²². In MUF-93, the $[\text{Zn}_4\text{O}(\text{L1})_3]$ sublattice templates the formation of $[\text{Co}_4\text{O}(\text{bpdc})_3]$ under secondary growth conditions. The emergence of the $[\text{Co}_4\text{O}(\text{bpdc})_3]$ sublattice is evidenced by SCXRD (Fig. 2d), ^1H NMR spectroscopy and atomic adsorption (AA) spectroscopy (which measured the Zn/Co ratio). As deduced by X-ray diffraction, large crystals of MUF-93 reach PIP levels of 70% over 60 hours of secondary growth (Fig. 1e and Supplementary Table 4), while starting from microcrystalline α -MUF-9 allows the $[\text{Co}_4\text{O}(\text{bpdc})_3]$ framework to reach 80% occupancy over just 18 hours (Supplementary Fig. 16). These PIP values align with the ^1H NMR and AA spectroscopic results over the early stages of secondary growth (Fig. 1e and Supplementary Fig. 14), which indicates that there is little displacement of the zinc(II) ions by cobalt(II). This is corroborated by the site-specific anomalous scattering experiments detailed later. While the exchange of three out of four zinc(II) ions per node is possible by heating α -MUF-9 in highly concentrated solutions of cobalt(II) nitrate

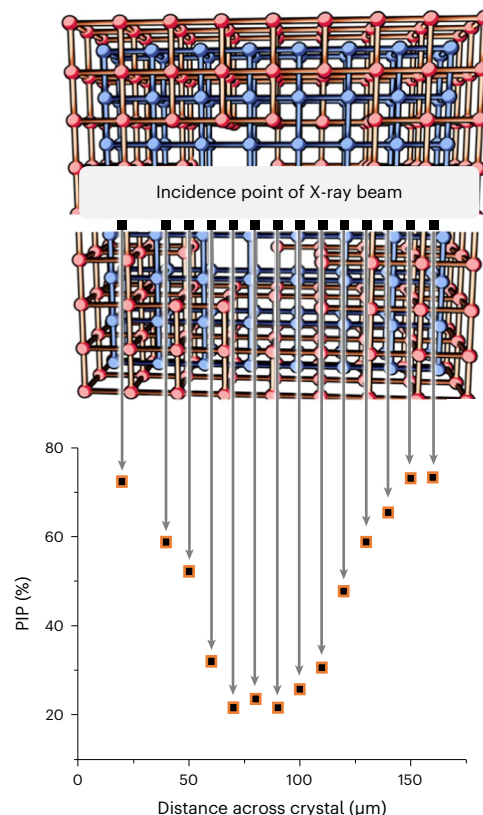


Fig. 3 | The variation in PIP across an individual crystal of MUF-93 as determined from multiple SCXRD datasets. An illustration of a crystal of MUF-93 is presented showing the regions pinpointed by the synchrotron X-ray beam. The occupancy level of the interpenetrating $[\text{Co}_4\text{O}(\text{bpdc})_3]$ sublattice is highest towards the edges of the crystal and lowest at its centre, consistent with secondary growth being more prevalent at the crystal surface than its centre. Note that the number of unit cells actually present in the X-ray beam at each data collection point is many orders of magnitude greater than illustrated.

in DBF (Supplementary Section 5.2), low rates of metal exchange during the secondary growth of MUF-93 were ensured by using a low concentration of cobalt(II) nitrate. Once the occupancy of the $[\text{Co}_4\text{O}(\text{bpdc})_3]$ sublattice reaches ~70% after 60 hours of secondary growth, the amounts of bpdc and Co rise beyond the PIP level due to the gradual displacement of the zinc(II) ions and **L1** linkers of $[\text{Zn}_4\text{O}(\text{L1})_3]$ by cobalt(II) and bpdc, respectively (Fig. 1e).

The spatial variation of PIP level within an individual MUF-93 specimen was probed by systematically collecting SCXRD datasets across the midpoint of a single crystal. The crystal was approximately ~180 μm in size, and synchrotron X-ray radiation with a spot size of approximately 10 μm (full-width at half-maximum) horizontally was used (Fig. 3 and Supplementary Fig. 13). Rastering of the crystal produced datasets from various spatial regions. As anticipated, we observed the highest PIP values (~72%) near both edges of the crystal since these regions are more accessible to the incoming components of the interpenetrating sublattice. The PIP levels drop to ~20% when the beam is directed at the centre of the crystal since the mass transfer limits the growth of the second lattice nearer the core.

Conventional single-crystal diffraction cannot reliably distinguish cobalt from zinc in MUF-93 because of their similar electron counts. This is exacerbated at low PIP levels due to high correlations between scattering factors, occupancy and atomic displacement parameters. However, tuning the X-ray wavelength to be near the respective absorption edges for cobalt and zinc enhances their anomalous dispersion and, in principle, permits the discrimination and quantification of these

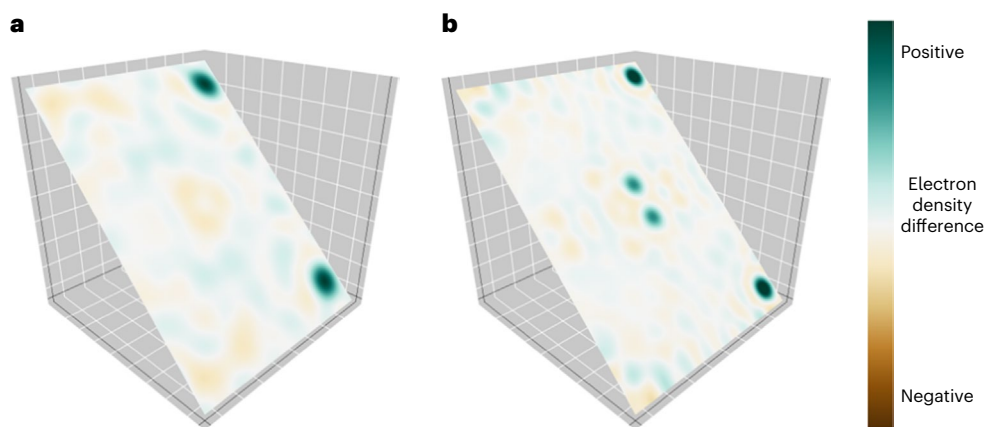


Fig. 4 | The differentiation of cobalt and zinc sites in hetero-interpenetrated MOFs using the anomalous dispersion of X-rays. Slices of the (101) plane of the anomalous electron density difference maps derived from X-ray diffraction datasets collected at 17,440 eV and 7,500 eV are presented. Peaks near the midpoint of the unit cell correspond to cobalt occupying the erstwhile Zn_4O cluster sites of the host sublattice, and peaks near the corner of the unit cell correspond to cobalt occupying Co_4O cluster sites of the interpenetrating

sublattice. **a**, MUF-93 after 60 hours of secondary growth showing cobalt(II) exclusively occupying the M_4O cluster sites of the interpenetrating $[\text{Co}_4\text{O}(\text{bpdc})_3]$ sublattice; a negligible quantity of it displaces zinc(II) from the $[\text{Zn}_4\text{O}(\text{L1})_3]$ host sublattice. **b**, MUF-93 after 168 hours of secondary growth showing that only after this prolonged reaction time does a detectable amount of cobalt(II) occupies the M_4O cluster sites of both sublattices. In this case, it has displaced some of the zinc(II) ions from the host $[\text{Zn}_4\text{O}(\text{L1})_3]$ sublattice.

metals at specific crystallographic sites (Supplementary Fig. 18)^{23–25}. On this basis, we developed a method for using anomalous dispersion to differentiate metals in crystalline materials. For MUF-93, we calculated the differences in reflection intensities between datasets collected just below the cobalt(II) absorption edge at 7,500 eV (where the in-phase anomalous scattering contribution (f') by cobalt is significant) and a high-resolution dataset collected at 17,440 eV (where there is little anomalous scattering) on the same crystal. The reflection intensity differences obtained in this way arise partly from the difference in anomalous scattering by the cobalt and thus can be used to locate and quantify the cobalt sites. Similarly, we used 9,670/17,440 eV difference datasets, which maximize zinc anomalous dispersion, to pinpoint the zinc sites. Figure 4 illustrates the datasets obtained in this way, showing differences between peaks near the unit cell centre (crystallographically identical metal atom sites of the $[\text{Zn}_4\text{O}(\text{L1})_3]$ host sublattice) and peaks near the unit cell corners (crystallographically identical metal atom sites of the interpenetrating $[\text{Co}_4\text{O}(\text{bpdc})_3]$ sublattice). After a secondary growth time of 60 hours, the occupancy of $[\text{Co}_4\text{O}(\text{bpdc})_3]$ in MUF-93 reaches its maximum. At this point, a distinct peak for cobalt appears near the corner of the unit cell in the difference datasets due to the cobalt ion in the $[\text{Co}_4\text{O}(\text{bpdc})_3]$ sublattice (Fig. 4a). No signal for cobalt can be detected near the midpoint of the unit cell, which demonstrates that cobalt(II) ions do not displace zinc(II) ions from the $[\text{Zn}_4\text{O}(\text{L1})_3]$ host over the 60 hours of secondary growth. Displacement of the zinc(II) ions in the $[\text{Zn}_4\text{O}(\text{L1})_3]$ sublattice becomes evident only after a much longer reaction time (Fig. 4b).

Two illustrative control experiments were also performed. First, a 7,500/17,440 eV difference dataset on a crystal of α -MUF-9 in which the zinc(II) ions had been partially replaced by cobalt(II) showed a single peak for cobalt near the midpoint of the unit cell and no sign of an interpenetrating lattice (Supplementary Fig. 19c). Second, as expected, a 9,670/17,440 eV difference dataset for homo-interpenetrated β -MUF-9 revealed two equally strong peaks for the two independent zinc sites in the unit cell ($P43m$ space group assumed; Supplementary Fig. 19d).

The anomalous dispersion experiments clearly show that the two sublattices in MUF-93 are distinct from each other, and thus the framework can be genuinely described as being hetero-interpenetrated. This observation underscores the power of the two-step strategy employing secondary growth since any attempt to synthesize MUF-93 directly

from a mixture of H_2bpdc , $\text{Zn}(\text{NO}_3)_2$ and $\text{Co}(\text{NO}_3)_2$ would result in a mixed-metal multivariate material.

(R)-MUF-101 and (S)-MUF-101 as catalysts

Hetero-interpenetrating frameworks constitute a design blueprint for unique functional properties. To illustrate this, we installed the two key roles of an asymmetric catalyst—catalytic activity and chirality—onto the two different sublattices of a hetero-interpenetrated MOF. In contrast to α -MUF-9, which comprises racemic **L1**, α -MUF-10 is a non-interpenetrated, chiral MOF built from enantiopure (*R*)-**L1** or (*S*)-**L1** (Fig. 5). The α -MUF-10 acts as the host sublattice, and its chiral pore space confers asymmetry on the catalysed transformations. An achiral interpenetrating $[\text{Zn}_4\text{O}(\text{L2})_3]$ sublattice delivers the catalytic activity. We selected **L2** to generate this sublattice since secondary amines are known to be effective organocatalysts for a range of reactions^{26,27}. Experimentally, α -(*R*)-MUF-10 or α -(*S*)-MUF-10 ($[\text{Zn}_4\text{O}((\text{R})\text{-L1})_3]$ or $[\text{Zn}_4\text{O}((\text{S})\text{-L1})_3]$) templates the growth of $[\text{Zn}_4\text{O}(\text{L2})_3]$ to produce MUF-101 (Fig. 5). For the catalysis studies, we employed MUF-101 with a PIP level of around 7% to retain high porosity and maximize the diffusion of the substrates and products in and out of the pores.

With a catalyst loading of around 3% (defined as the molar ratio of catalytic **L2** units to aldehyde), MUF-101 catalyses the aldol reaction between 2-chloro-5-nitrobenzaldehyde (**1**) and acetone (Fig. 5). High-performance liquid chromatography (HPLC) revealed the reaction product (**2**) to have an enantiomeric excess (e.e.) of -13.9% when catalysed by (*R*)-MUF-101 (Supplementary Fig. 22 and Supplementary Table 7). Importantly, the enantioselectivity was reversed when (*S*)-MUF-101 was used as the catalyst (Supplementary Table 8). This confirms that the preferred handedness of the reaction product is induced by the chirality of the MUF-10 host sublattice. To verify this, we grew a $[\text{Zn}_4\text{O}(\text{L2})_3]$ sublattice inside the racemic α -MUF-9 host to produce the MUF-99 framework. As expected, MUF-99 is catalytically active but does not confer asymmetry on **2** due to the racemic host environment. As an additional test of the activity of MUF-101, we found that it catalyses the Henry reaction between **1** and nitromethane (Supplementary Scheme 4). The reaction product (**3**) has an e.e. of -9.4% when catalysed by (*R*)-MUF-101, the sign of which is reversed when (*S*)-MUF-101 is the host sublattice (Supplementary Table 8).

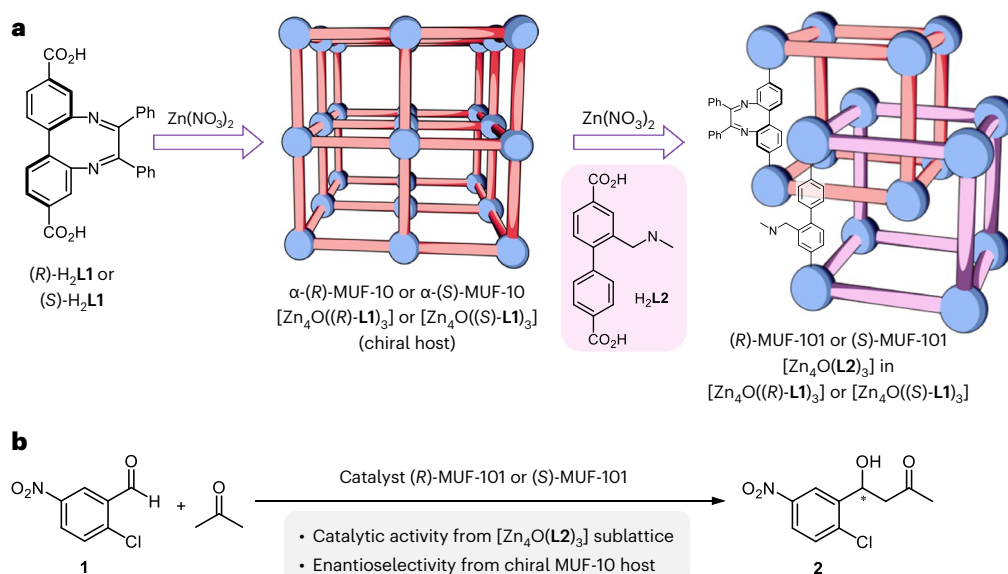


Fig. 5 | Hetero-interpenetrated MOFs as asymmetric catalysts. In MUF-101, chiral $\alpha\text{-(R)-MUF-10}$ or $\alpha\text{-(S)-MUF-10}$ hosts a catalytically active, yet achiral, $[\text{Zn}_4\text{O}(\text{L2})_3]$ sublattice. MUF-101 acts as an asymmetric catalyst by coupling the chirality of the host to the catalytic activity of the interpenetrating sublattice. **a**, The synthetic route to MUF-101. **b**, The aldol reaction between 2-chloro-

5-nitrobenzaldehyde (**1**) and acetone is catalysed by MUF-101 and produces **2**. The e.e. of **2** changes sign depending on whether $\alpha\text{-(R)-MUF-10}$ or $\alpha\text{-(S)-MUF-10}$ is used as the host sublattice since this lattice confers asymmetry on the reaction process.

Outlook

The rational design of hetero-interpenetrated MOFs has inherent challenges since the component sublattices must be geometrically and chemically compatible, phase separation is possible and the exchange of the metal ions and linkers may occur. Here we have taken advantage of the templating effect of a host lattice to promote the growth of interpenetrating sublattices in a second, discrete step. These hetero-interpenetrated MOFs retain the make-up of the individual sublattices since there is no detectable exchange of components between them. In certain cases, methods to produce the interpenetrating sublattices have not been previously reported.

We are optimistic that this deliberate strategy regarding hetero-interpenetrated MOFs will open new perspectives on the field of framework chemistry. Specific functional properties can emerge when different and complementary sublattices are coupled to one another. Here we show how the two principal roles of an asymmetric catalyst can be assigned to the different sublattices: the host framework provides a chiral environment for the second, catalytically active, sublattice. The enantioselectivity is dictated by the handedness of the host sublattice in a way that is reminiscent of the active site in enzymes where the chirality of the catalytic pocket influences the reaction stereochemistry.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41557-023-01277-z>.

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Data availability

Crystallographic data for the structures reported in this Article have been deposited at the Cambridge Crystallographic Data Centre, under deposition numbers [2144458–2144473](#) and [2269651](#) (MUF-91), [2144574–2144588](#) (MUF-92), [2149445–2149471](#) and [2269651](#) (MUF-93) and [2149424–2149437](#) (rastering files for MUF-93), as detailed in Supplementary Table 1. Copies of the data can be obtained free of charge via <https://www.ccdc.cam.ac.uk/structures/>. All other data are included in the Supplementary Information. Source data are provided with this article and also available on figshare, <https://doi.org/10.6084/m9.figshare.22682965.v2> (ref. 28).

Code availability

The Python scripts are available on figshare at <https://doi.org/10.6084/m9.figshare.22682965.v2> (ref. 28).

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Author contributions

S.G.T. conceived the idea. D.P. developed the concept, conducted experiments, analysed the results and prepared the Supplementary Information. A.F. and S.J.L. conducted experimental work. G.B.J. conceived and advised on the anomalous scattering experiments. S.G.T. and G.B.J. supervised the project. S.G.T. and D.P. wrote the manuscript with contributions from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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